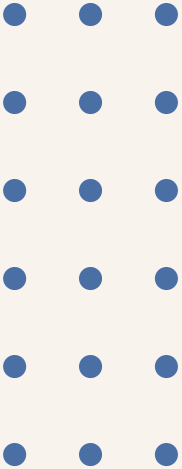


~~Raised~~ Expectations

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RESEARCH FRAMING

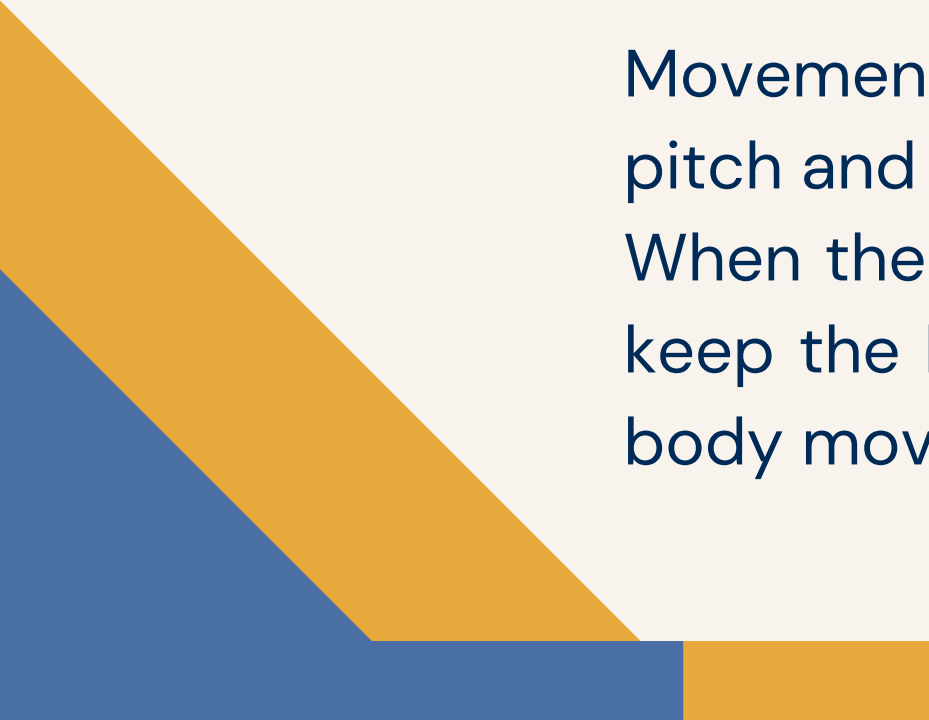
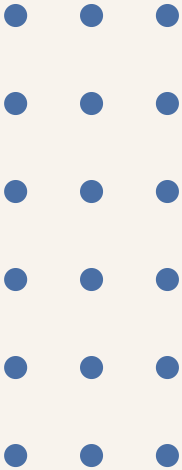


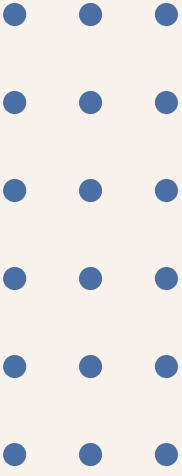
The social dynamic examined was interpersonal communication in situations of reduced biomechanical stability, which is both societally and scientifically significant.

Most previous research on aspects of vocal production have been in stable lab environments, often in seated positions (Zsiga, 2014). However, this doesn't consider that humans are often not stationary.

Movement, such as walking, can have an impact on aspects of the voice, such as pitch and rhythm (e.g., Anikin, 2023; Chiu et al., 2026).

When the body is moving, several biological mechanisms are employed in order to keep the body upright, with the body's centre of gravity constantly shifting as the body moves (Tesio & Rota, 2019).





WHY PITCH AND EYEBROWS?

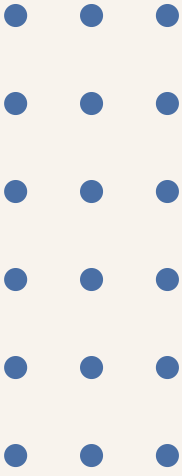


In English, pitch plays important prosodic signalling functions at both the word level (e.g. stress) and the sentence level (e.g. intonation and information structure).

There is evidence that eyebrow raises may serve a similar prosodic function in the visual domain, marking prominence and conveying communicative functions such as turn-taking (Cavé, Guaïtella & Santi, 2002).

Perception research suggests that when pitch information provides conflicting cues to prominence, people will use eyebrow information to infer the location of prominence (Krahmer et al., 2002).

Still unclear exactly how well coordinated pitch and eyebrow movements are, and how eyebrow movements may coordinate between interlocutors.



RESEARCH QUESTIONS



Q1: What is the relationship between pitch and eyebrow movements?
Q1.2: How does this relationship change, if at all, when the bodily center of gravity is perturbed?



RQ2: What is the nature of the eyebrow synchrony between the clue giver and guesser?
RQ2.2: How is this changed, if at all, when the bodily centre of gravity is perturbed?

HYPOTHESES

- Raised eyebrows are associated with higher pitch in the clue-giver when they are standing on the ground.
- Under increased postural stability demands (the clue-giver is standing on the wobble board), the co-variation between eyebrow movement and pitch dynamics is distorted.
- Higher mutual information in eyebrow movements between the clue giver and the guesser is associated with faster guessing times (higher task performance).

METHODOLOGY

BALANCE DATASET

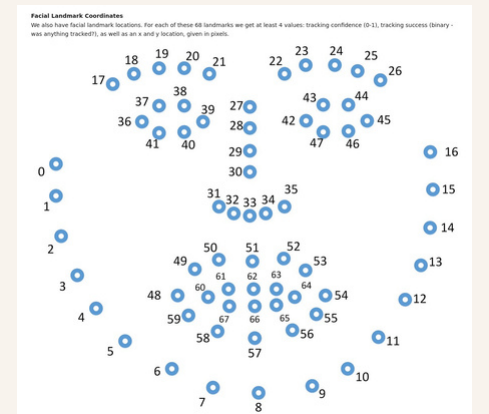
(LI ET AL., 2026)

Two dyads (f=3, m=1, mean age= 19.5)

One target word ('Birthday')



Eyebrow – OpenFace data extracted from video data. Took the X and Y coordinates for the eyebrows (points 17–26) and the tip of the nose (point 33). Relative positions of points 17–26 as well as y-axis displacement of eyebrow centre of mass from tip of nose. *Both guesser and cluegiver*



Pitch – Extracting fundamental frequency (f0) for audio files using the parselmouth library [jadoul_etal18]. We used the Diarization data to split the f0 by speaker. This was manually checked for the examples (for a larger analysis pipeline, we plan to use a multi speaker separation as this would allow potential extraction of pitch information when two speakers are speaking simultaneously. *Cluegiver only*)

METHODOLOGY

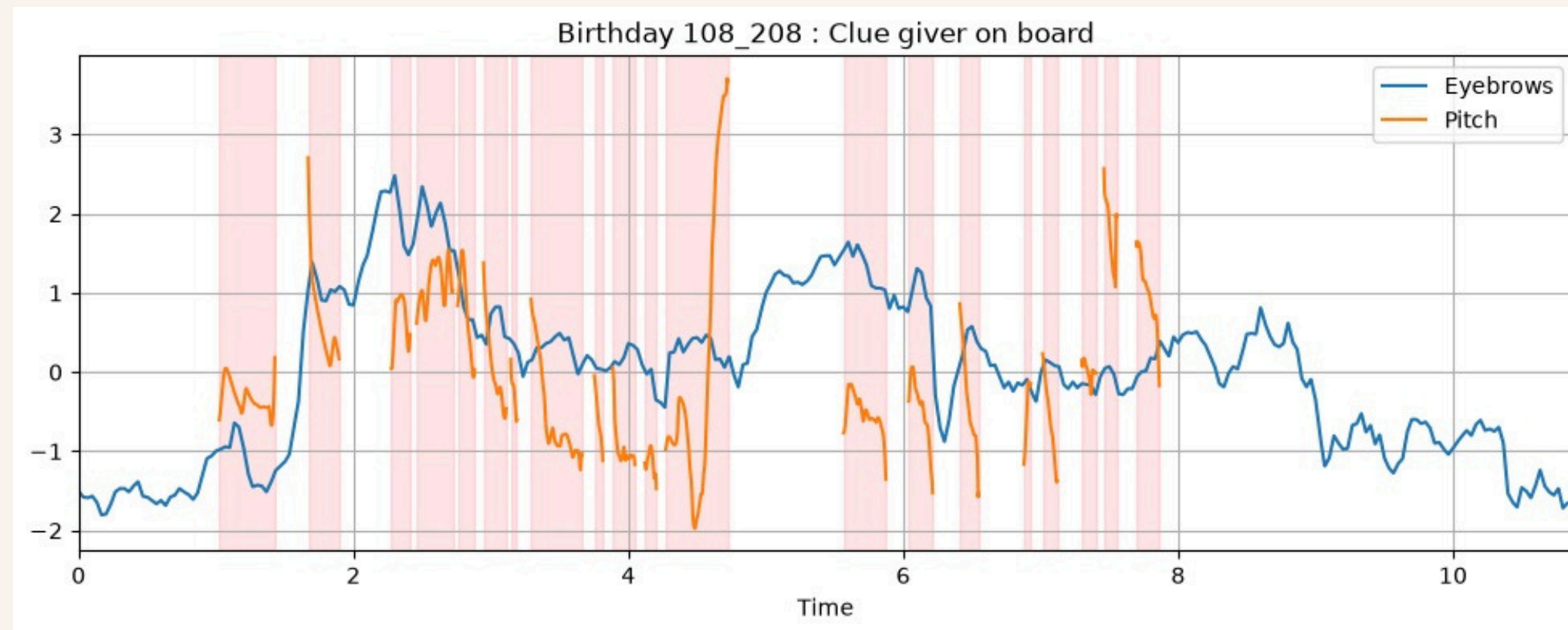
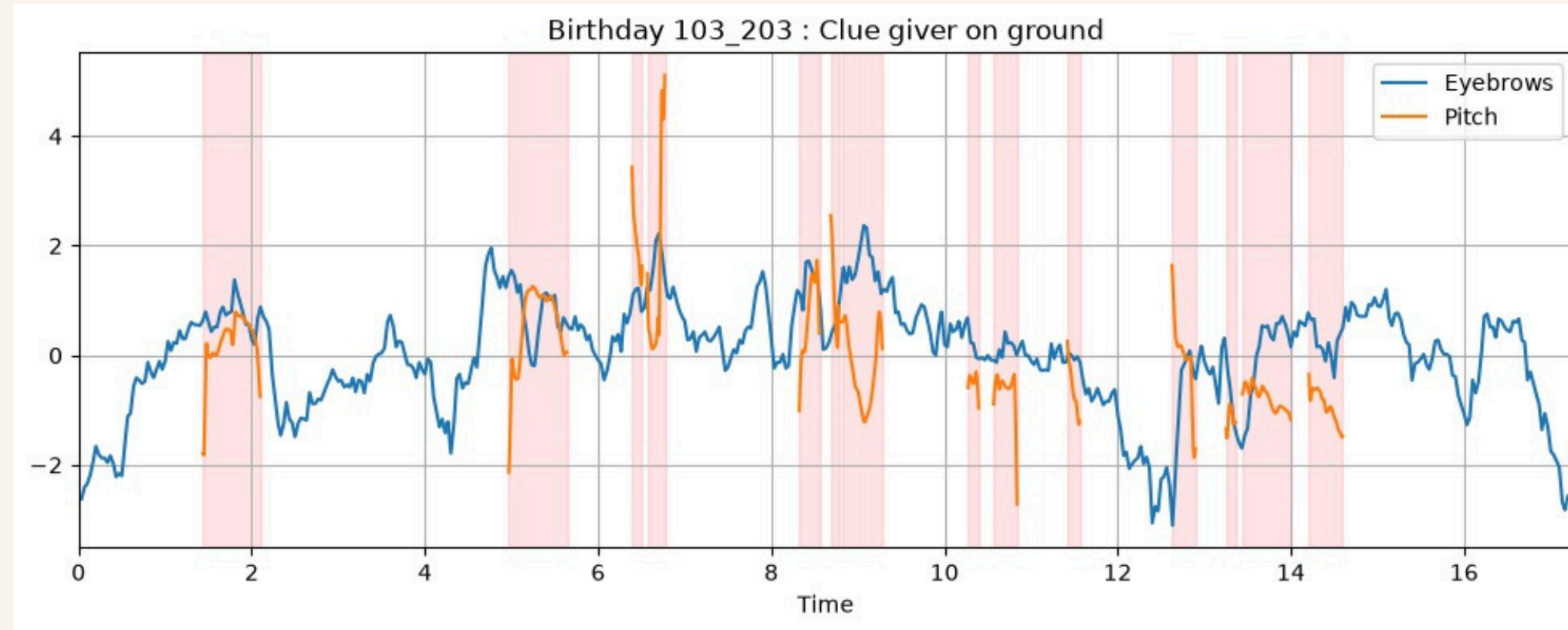
ANALYSIS

Hypothesis 1: Linear and non-linear interaction measures: correlation, covariance, mutual information, time-delayed mutual information (JIDT + pySPI); WMS

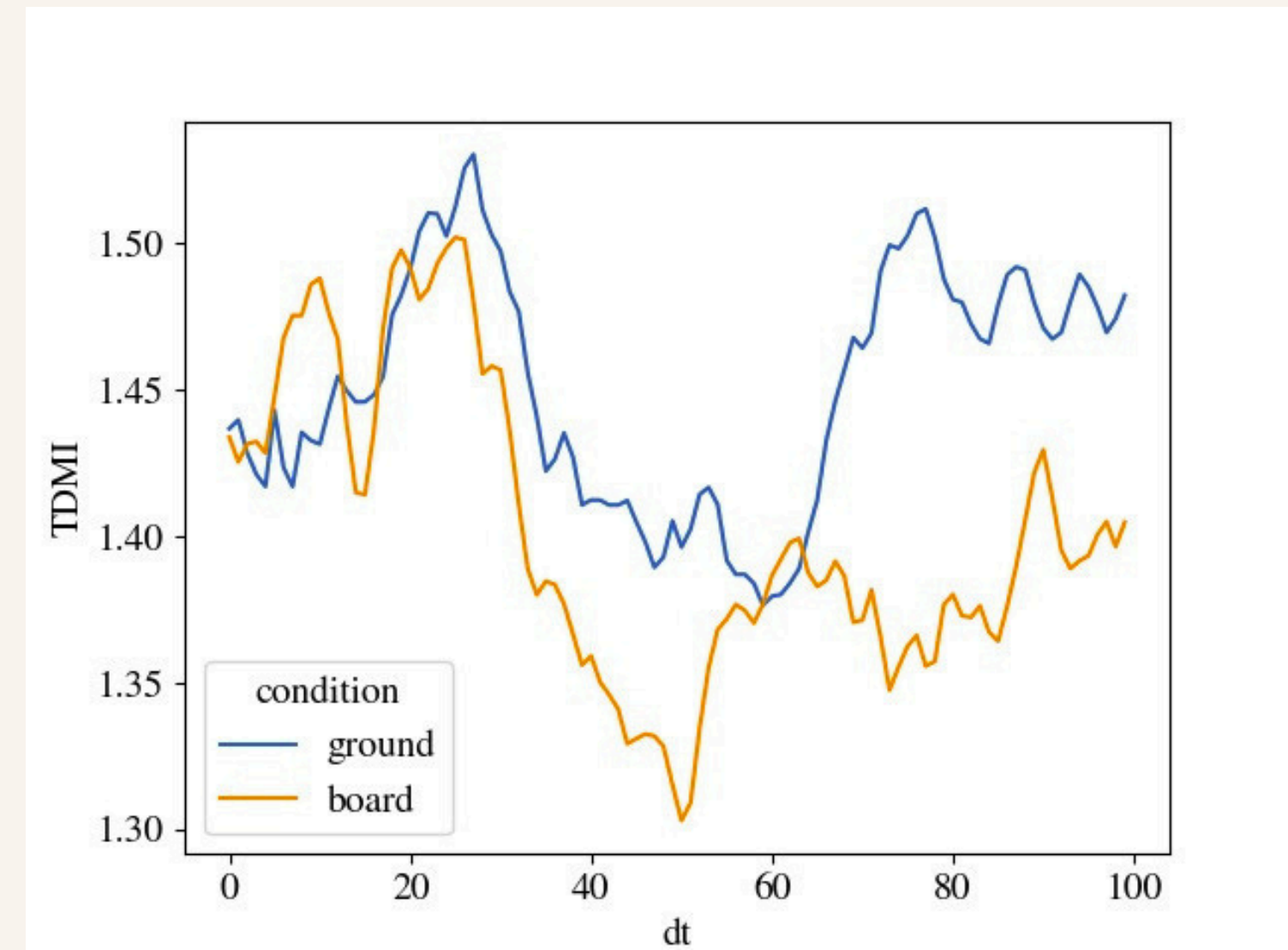
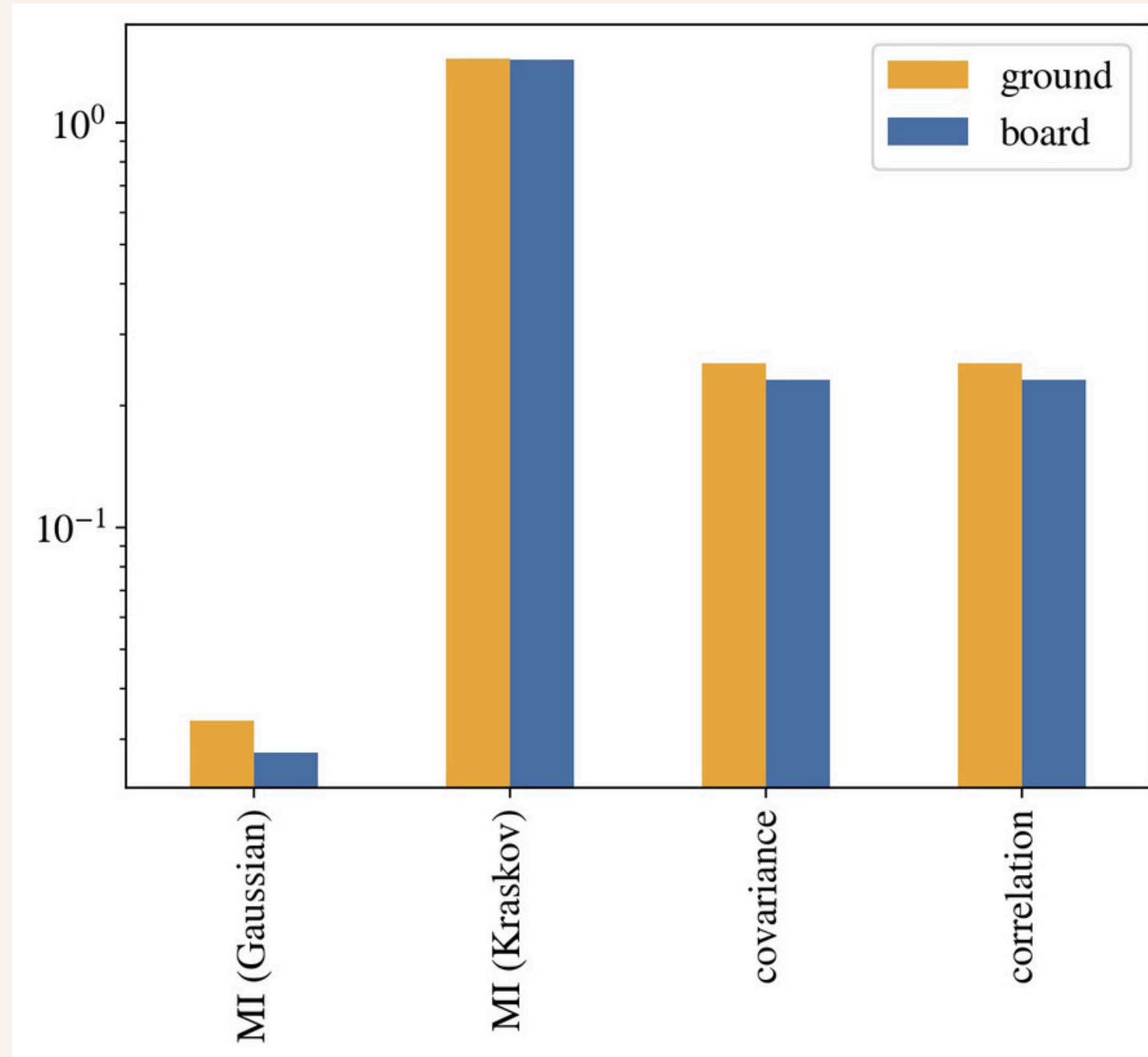
Hypothesis 2: Cross-Recurrence Quantification Analysis with phase-space reconstruction to analyse coupling between eyebrow movement and pitch dynamics on an intrapersonal level

Hypothesis 3: Multivariate mutual information applied to brow positions (JIDT)

WINDOWED MULTISCALE SYNCHRONY

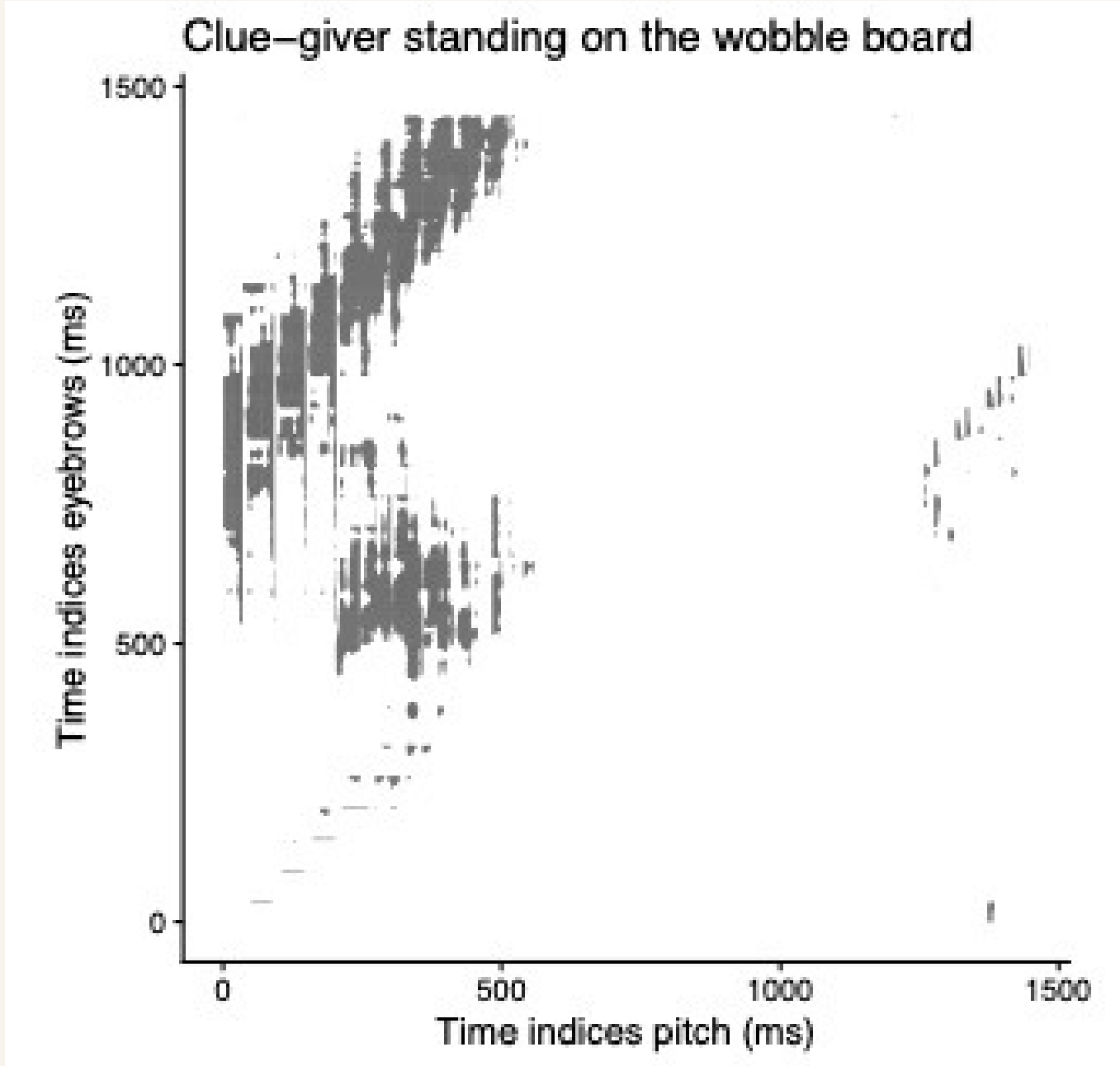
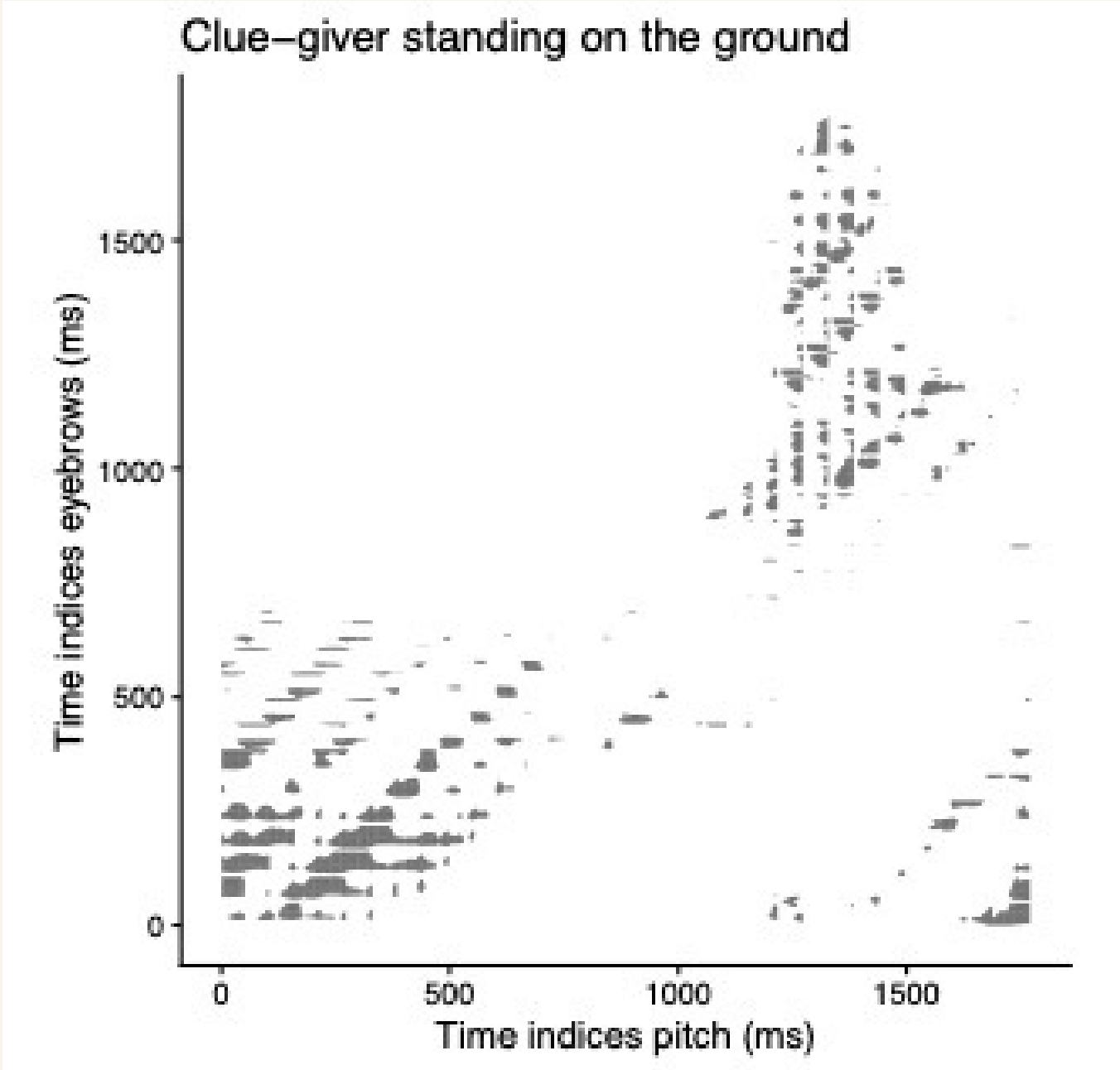


TIME-DELAYED MUTUAL INFORMATION



CROSS-RECURRENCE QUANTIFICATION ANALYSIS

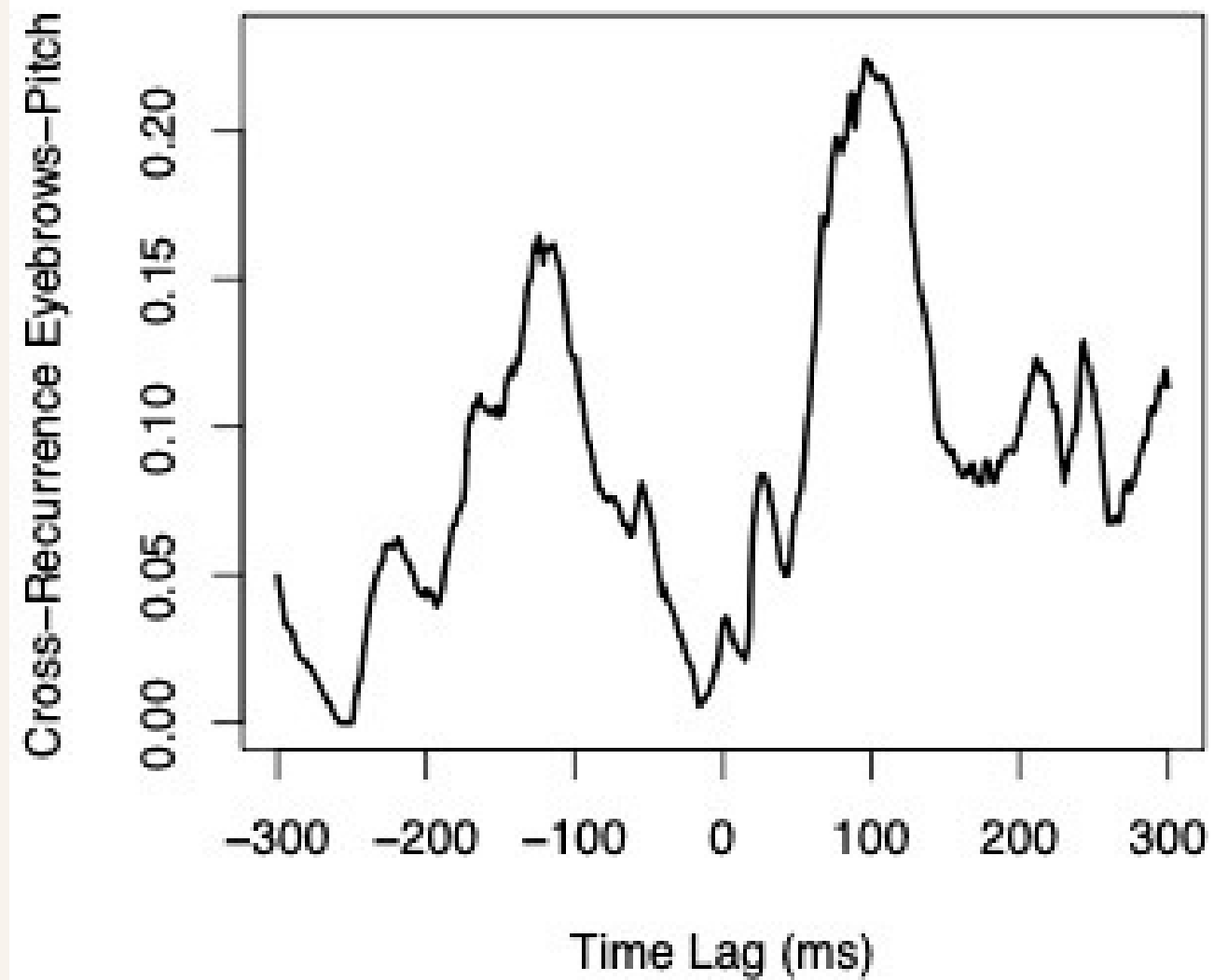
CRQA
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embed = 8
radius = 2



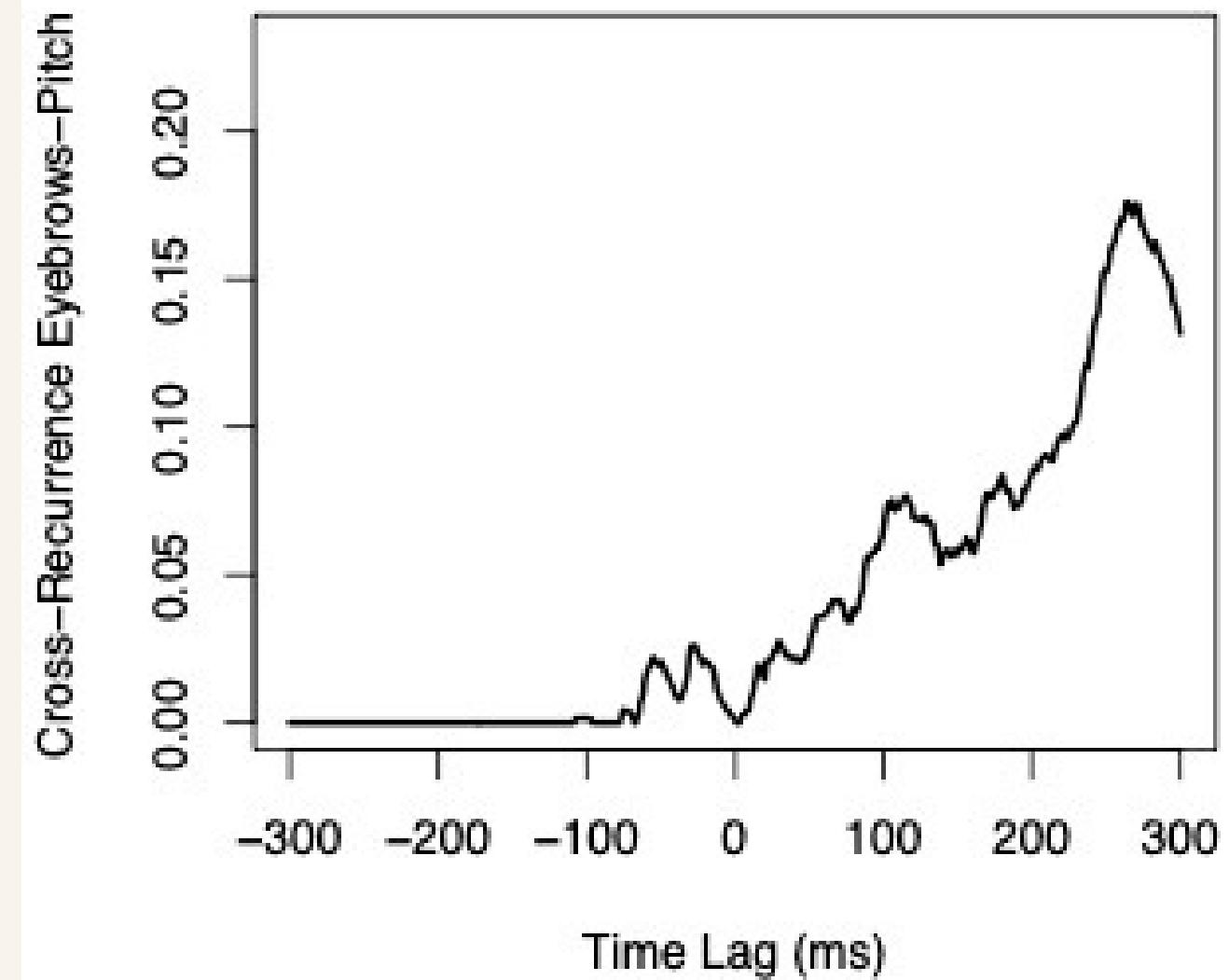
condition	RR	DET	L	rENTR	LAM	TT
ground	4.19387787672763	99.4603851130223	13.2603754439356	0.882565486246514	99.8652865514879	28.4072310023815
wobble_board	7.63000522259144	99.3619035082771	18.4517518333128	0.766883648908213	99.5298896180823	21.1377795527157

DIAGONAL CROSS-RECURRENCE PROFILE

Clue-giver standing on the ground



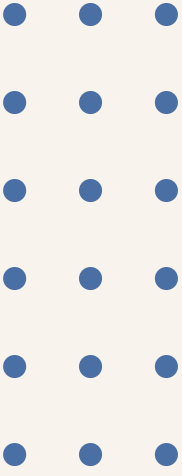
Clue-giver standing on the wobble board





SUMMARY

Some preliminary evidence (limited) that supports differences between standing on wobble board and standing on ground for how eyebrow movement and pitch are coordinated. Further analysis necessary.



CHALLENGES AND FUTURE DIRECTIONS

Number of datapoints is not the same for all trials in the dataset (and not many datapoints at that)

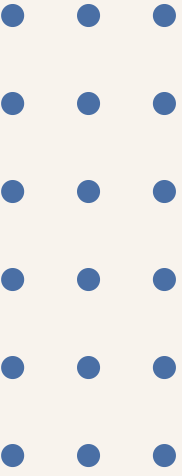
Head gear – presence of helmet can restrict eyebrow and facial movements

Botulinum Toxin (e.g., Botox) and other muscular argumentation.

- Eyebrows in particular susceptible to freezing and reduced movement due as a result of popular aesthetic procedures (Schlager et al., 2019)

Languages play a large role in pitch production, so can expect a cross-linguistic difference in relationship between eyebrow movement and pitch

Extracting accurate eyebrow movement in moving head systems. Accounting for head roll, etc.



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